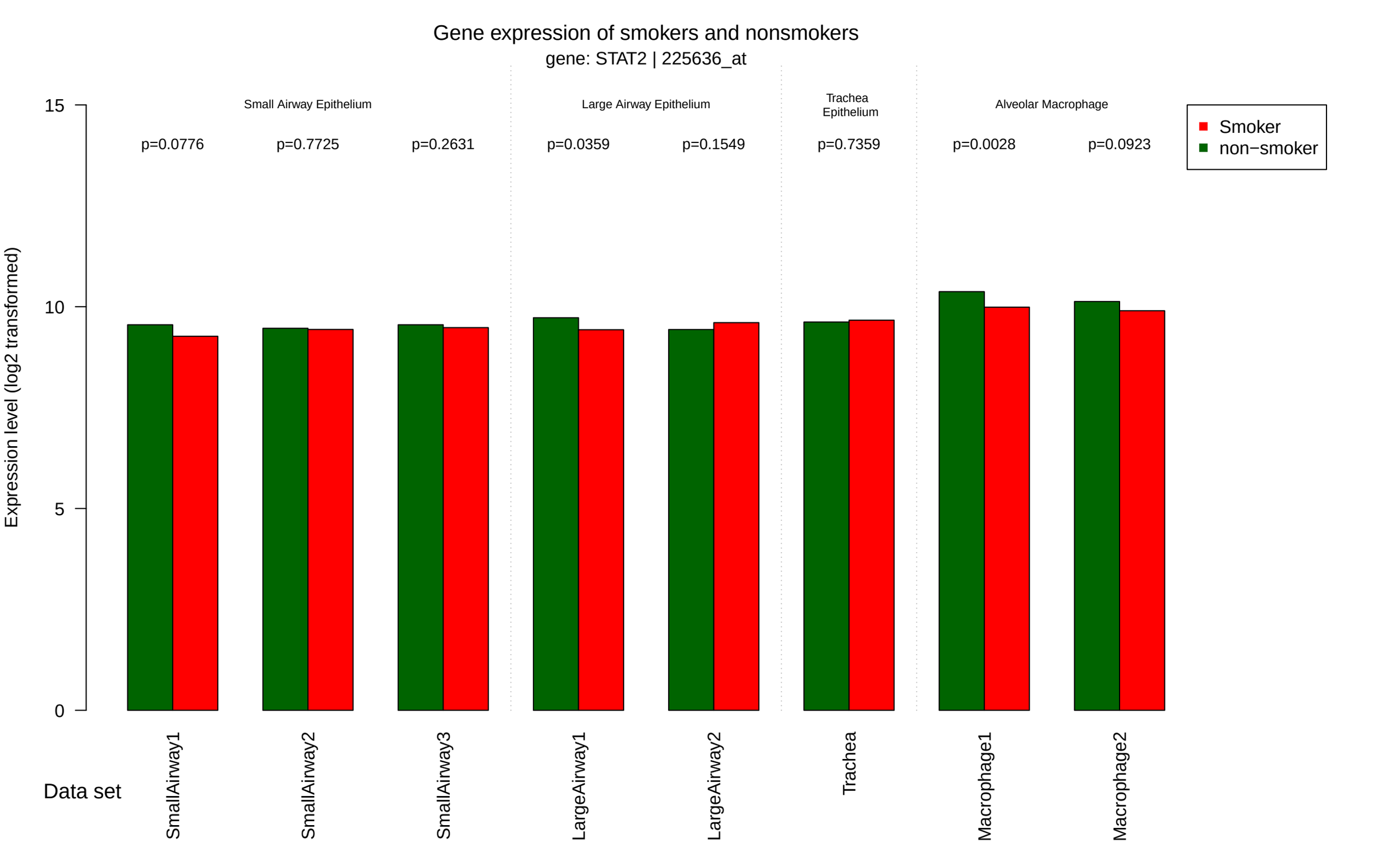
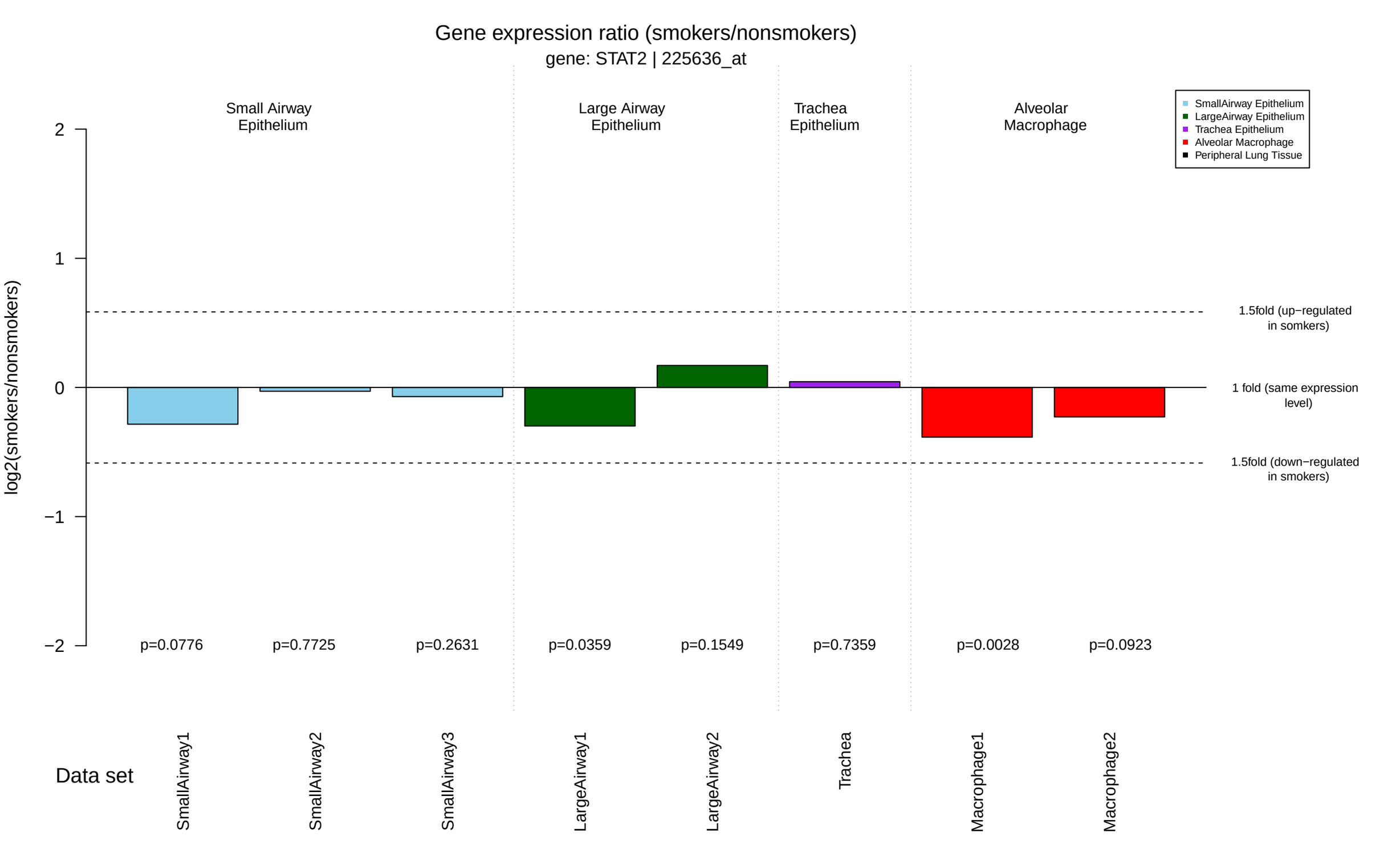
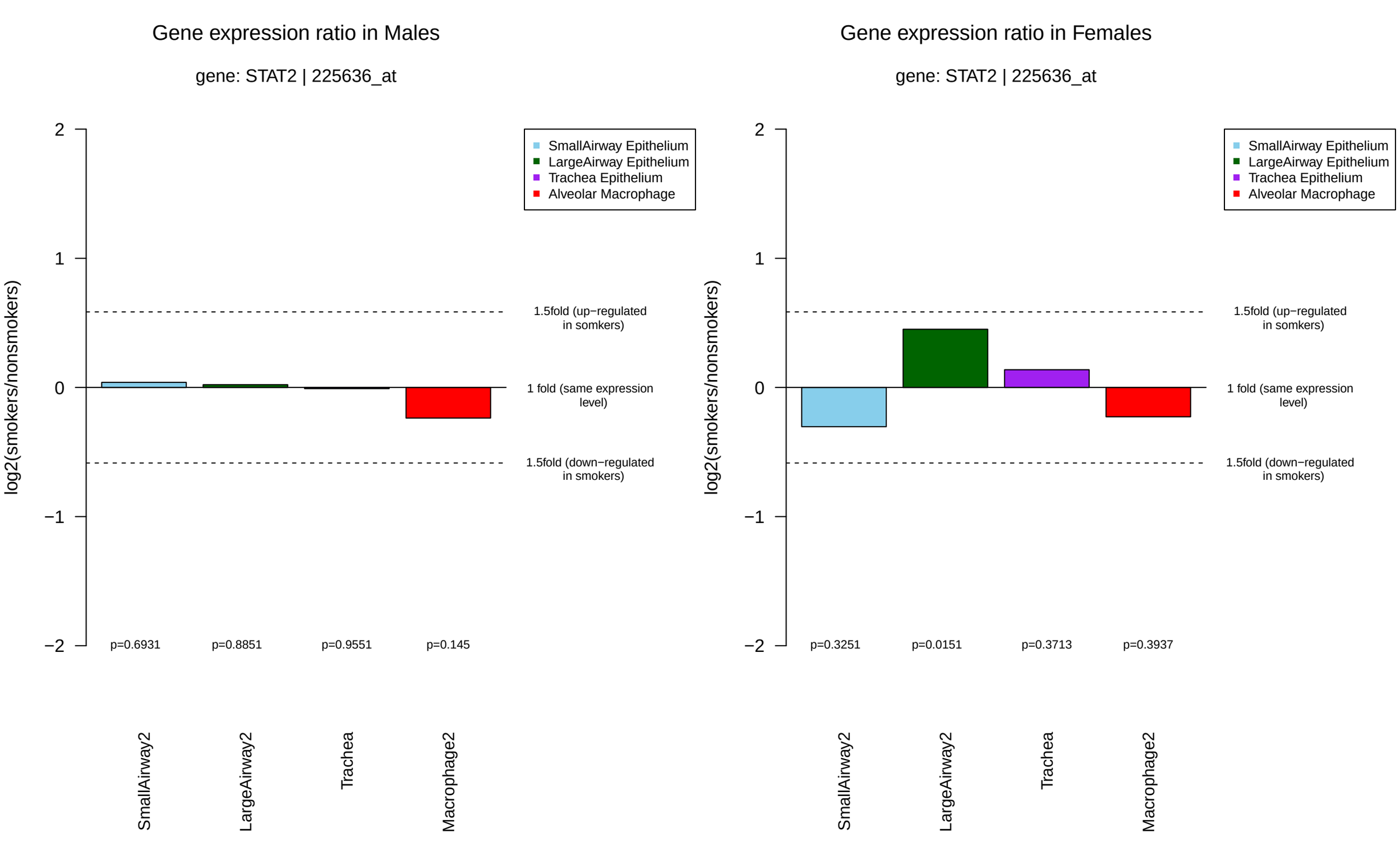
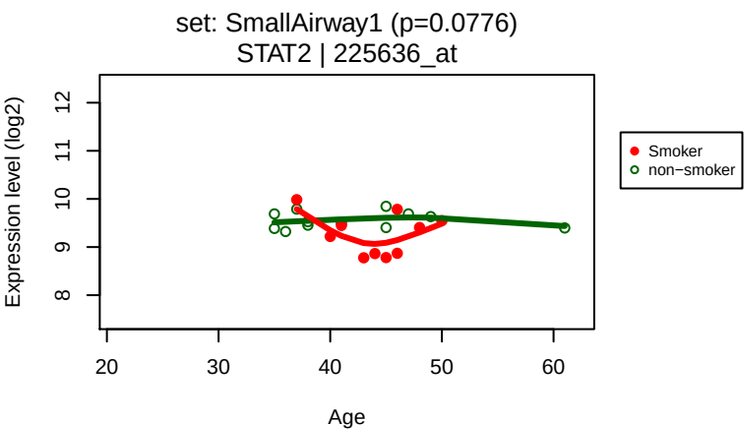
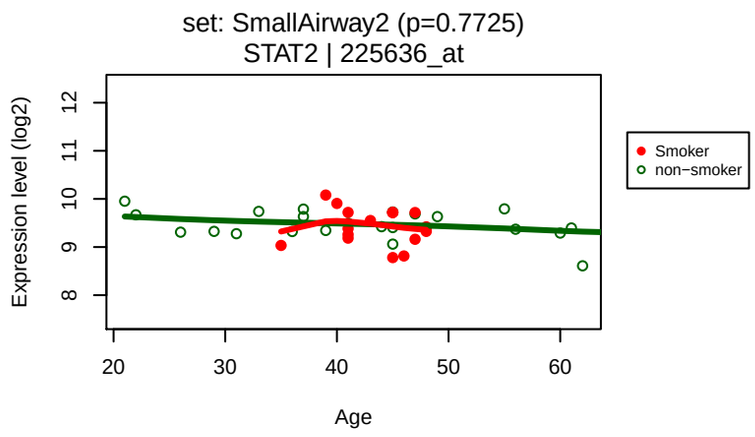
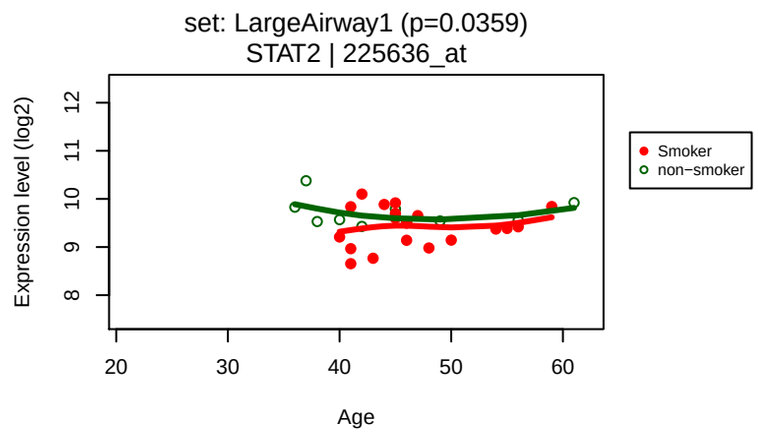

Gene expression of smokers and nonsmokers
gene: STAT2 | 225636_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0776
p=0.7725
p=0.2631
p=0.0359
p=0.1549
p=0.7359
p=0.0028
p=0.0923
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

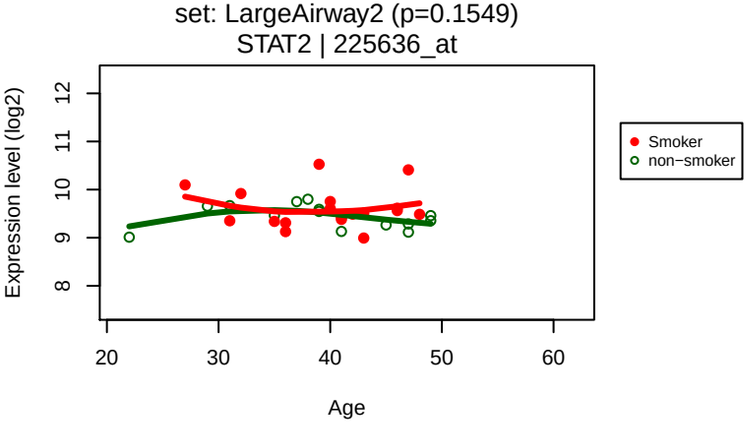

Gene expression ratio (smokers/nonsmokers)
gene: STAT2 | 225636_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0776
p=0.7725
p=0.2631
p=0.0359
p=0.1549
p=0.7359
p=0.0028
p=0.0923
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

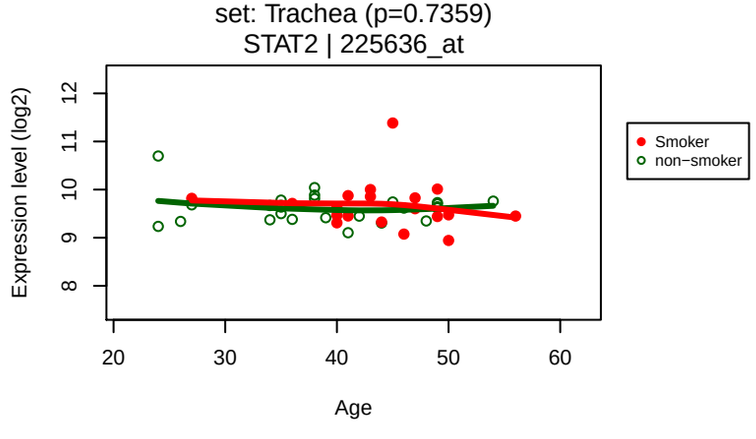

Gene expression ratio in Males
gene: STAT2 | 225636_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6931
p=0.8851
p=0.9551
p=0.145
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STAT2 | 225636_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3251
p=0.0151
p=0.3713
p=0.3937
SmallAirway2
LargeAirway2
Trachea
Macrophage2

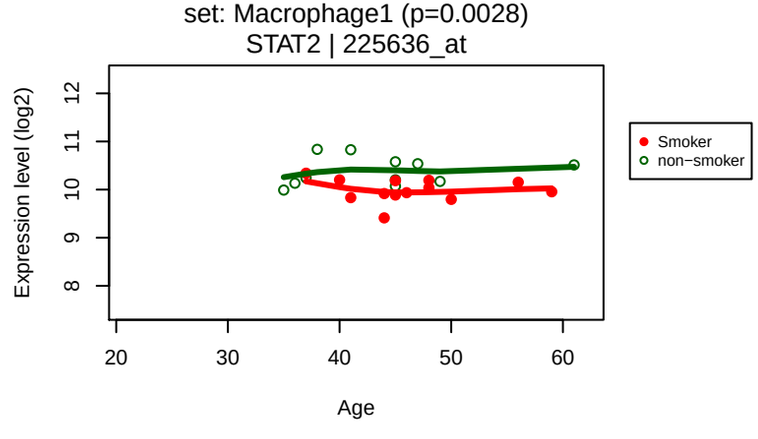

set: SmallAirway1 (p=0.0776)
STAT2 | 225636_at
Expression level (log2)
Age
Smoker
non−smoker

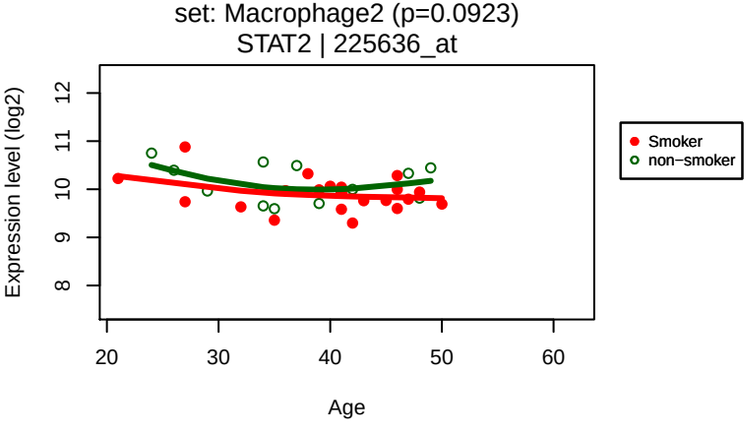

set: SmallAirway2 (p=0.7725)
STAT2 | 225636_at
Expression level (log2)
Age
Smoker
non−smoker

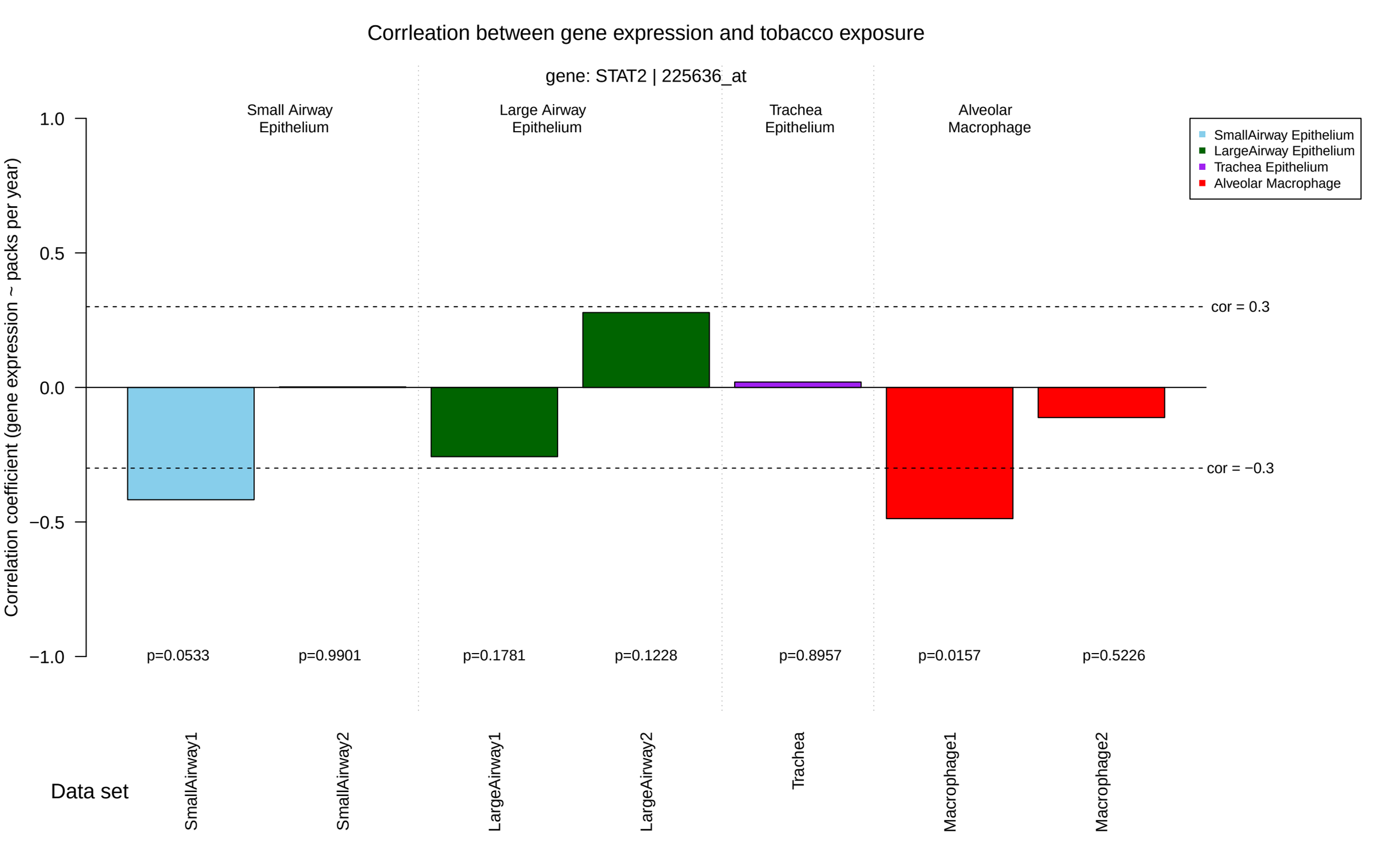

set: LargeAirway1 (p=0.0359)
STAT2 | 225636_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1549)
STAT2 | 225636_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7359)
STAT2 | 225636_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0028)
STAT2 | 225636_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0923)
STAT2 | 225636_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STAT2 | 225636_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0533
p=0.9901
p=0.1781
p=0.1228
p=0.8957
p=0.0157
p=0.5226
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2